

JOHN MARK GATCHE

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SUMMARY

Computer Science graduate and DOST-SEI Scholar from the University of the East – Manila (Summa Cum Laude, 1.08 GWA) with full-stack development experience: frontend (React, TypeScript, Next.js, Tailwind CSS) and backend (Python, FastAPI, Node.js, Laravel), building applications backed by relational databases (Microsoft SQL Server, PostgreSQL, SQLite). Additional experience in Android development with Java, Kotlin and Jetpack Compose, and AI/ML solutions using Python, including traditional ML and deep learning techniques. Particularly interested in software engineering, UI/UX, and applied AI/ML.

SKILLS

Languages:	Python, JavaScript, TypeScript, SQL, C++, C#, Java, Kotlin, LaTeX, PHP, HTML, CSS
Libraries & Frameworks:	React.js, Next.js, Tailwind CSS, FastAPI, Jetpack Compose
Data Science & ML:	Jupyter Notebook, RStudio, scikit-learn, pandas, NumPy
Design & Visualization:	Adobe Photoshop, Adobe Illustrator, Blender, Figma, Microsoft PowerPoint
Tools & Platforms:	Git, GitHub, Supabase, IntelliJ IDEA, PyCharm, WebStorm, Visual Studio Code, Node.js

EDUCATION

University of the East – Manila | *BS Computer Science*

Graduated June 2026

- DOST-SEI Scholar and University Scholar | GWA: 1.08 (Summa Cum Laude)

EXPERIENCE

DataAnnotation | *Freelance AI Trainer*

July 2026 – Present

- Evaluate AI-generated code for correctness, efficiency, readability, and adherence to software engineering best practices.
- Analyze programming solutions to identify bugs, edge cases, and opportunities for improvement.
- Produce clear, evidence-based technical evaluations while independently managing projects and consistently meeting quality standards.

Big Pond Education | *Frontend Developer Intern*

June 2025 – August 2025

- Developed responsive interfaces for a prototype information management system for Local Government Units (LGUs) with TypeScript, React.js, and Tailwind CSS and resolved database/API bugs in the Laravel and SQLite backend.
- Presented the prototype to LGUs, generating interest from local administrators as a potential solution for streamlining resident request processing.
- Continued post-internship work migrating the database to PostgreSQL.

LEADERSHIP & ACTIVITIES

Junior Philippine Computer Society – UE Manila Chapter | *Dept. Head of Software Development*

July 2025 – April 2026

- Led development of the organization's official website (jpcs-ue.tech).
- Collaborated with student officers to implement website updates supporting organizational initiatives.

UE CCSS Research & Development Unit | *Research Committee Assistant*

August 2023 – June 2024

- Coordinated and facilitated online cybersecurity webinars to enhance awareness among students.
- Assisted in planning and executing college week competitions promoting math and computer science.

PROJECTS

LakadPH – Pedestrian-First Smart Navigation Web App | *Hackathon Project*

March 2026

- A pedestrian-focused mobility platform that helps users find safer, cooler, and more comfortable walking routes in Metro Manila by ranking paths based on building shade coverage during the day and streetlight density at night.
- Built a responsive frontend prototype with TypeScript, React.js, and Tailwind CSS, integrating OpenStreetMap for map data, ShadeMap API for shade coverage, and Overpass API for streetlight location data.
- Won Champion (1st Place) at Blue Hacks 2026 Hackathon held at Ateneo de Manila University.

Pasada – AI-Powered Driving Optimization for Jeepney Drivers | *Hackathon Project*

August 2025

- Designed a progressive web app targeting Metro Manila's 75,000 jeepney and transport drivers, delivering real-time route intelligence, passenger demand heatmaps, and optimal timing recommendations to reduce wasted fuel and maximize daily take-home income.
- Built an offline-capable PWA with TypeScript, React, and Tailwind CSS using Vite and service workers, featuring Mapbox GL for live route mapping, voice-assisted fare collection via the Web Speech API, and trip analytics dashboards for earnings visibility.
- Advanced to Top 10 Finalist at the Byte Forward Hackathon 2025.

Vyve – AI-Powered Productivity Overlay for Filipino Freelancers | *Hackathon Project*

July 2025

- Designed an AI-powered productivity platform with a universal overlay that integrates across any software tool (Figma, VSCode, Notion), addressing the 9 app switches per hour and 65% PM tool abandonment rate experienced by 1.5M Filipino freelancers.
- Built AI assistant features including task auto-generation from natural language prompts, predictive workflow insights, persona-adjustable AI with contextual memory, and distraction-free creative overlays with tiered plans from free to enterprise with local LLM deployment.
- Advanced from 32 teams to Top 9 Finalists at Create & Conquer 2025 (FEU Tech); won the ZenithPads Award and pitched to judges from DOST-NCR, PwC Philippines, iCorePay, and DICT.

GabAI – SMS-Based AI Chatbot for Rural Barangay Governance | *Ideathon Project***March 2025 – April 2025**

- Engineered an offline-first AI chatbot architecture that processes local ML models to interface directly with SMS and USSD gateways, enabling citizens without internet access or smartphones to query verified government information, public programs, and health advisories.
- Designed a lightweight message-routing system and volunteer moderation dashboard to manage high-volume text queries securely to bridge the information gap for remote and underserved barangay communities across the Philippines.
- Won the Grand Prize in the Public Administration category at the HABI 2025 FLIP Workshop (2025 National Innovation Day), competing against 62 Higher Education Institutions nationwide.

Scaffl.ed – LLM-Driven Java Debugging Tutor via Gaze & Mouse Tracking | *Thesis Project***August 2024 – October 2025**

- An intelligent tutoring system for Java programming that detects student confusion in real-time using webcam-based eye tracking and mouse interaction data, providing adaptive scaffolded guidance through a 3D avatar robot tutor powered by Gemini 2.5 Pro.
- Led a five-member team managing project planning, research writing, and technical execution; built the full-stack web application using TypeScript/Next.js frontend and Supabase (PostgreSQL)/FastAPI backend with a 3D animated robot tutor avatar (Three.js/Blender) and real-time WebGazer.js eye tracking.
- Spearheaded the collection and curation of gaze/mouse interaction datasets and developed a Random Forest confusion detection model in Python using scikit-learn.
- Earned Best Thesis (96.2%) among 3rd-year Computer Science students at UE Manila; research project was shortlisted in the top 20% of over 1,600 global submissions at the QS Reimagine Education Awards.
- Co-authored and presented the research at the 2026 International Conference on Advances in AI and Machine Learning (AAIML) in Tokyo, Japan, receiving the Outstanding Oral Presentation Award; publication indexed in IEEE Xplore and Scopus.

Studdy – Study Partner Matching Application | *Startup Project***January 2024 – October 2024**

- Led development of an Android study partner matching application, architecting the data layer with a repository pattern and dependency injection, and building the UI with Jetpack Compose under an MVVM architecture.
- Built custom UI components using Material 3 and low-level Compose drawing APIs to achieve specific design requirements.
- Integrated the app with a pre-built ASP.NET backend via Retrofit, handling API requests, response parsing, and error handling to ensure reliable data sync with the UI.
- Achieved 2nd Runner-Up, 2024 Philippine Startup Challenge – NCR Regional Pitching Competition.

ACHIEVEMENTS**Blue Hacks 2026** | *1st Place***March 2026**

- Competed in a 25-team hackathon organized by CompSA at Ateneo de Manila University, with a focus on green urban cities and sustainable urban development.
- Built the LakadPH prototype (pedestrian-first shade-aware navigation web app) within the 24-hour development window and won 1st Place overall.

AAIML 2026 | *Outstanding Oral Presentation***March 2026**

- Co-authored and presented the Scaffl.ed research paper at the 2026 International Conference on Advances in Artificial Intelligence and Machine Learning (AAIML 2026) at Chuo University in Tokyo, Japan.
- Received the Outstanding Oral Presentation Award among global researchers; paper published and indexed in IEEE Xplore and Scopus ([DOI: 10.1109/AAIML67890.2026.11498122](https://doi.org/10.1109/AAIML67890.2026.11498122)).

14th UMak IT Skills Olympics | *Champion***November 2025**

- Won Champion in the IT Quiz Bee category at the 14th UMak IT Skills Olympics, a nationwide intercollegiate competition hosted by the University of Makati's College of Computing and Information Sciences.
- Tested across a broad range of ICT topics including OOP, data structures, networking, database systems, cybersecurity, data science, and software engineering against 40+ universities.

CodeChum National Programming Challenge 2025 | *3rd Place (Finals)***November 2025**

- Won 1st Place in Group Stage 2 and 1st Place in Semi-Finals before placing 3rd in the Finals, advancing to the national Grand Finals as one of the Group Stage Champions.
- Competed against dozens of partner institutions nationwide in timed algorithmic challenges on CodeChum's automated online judge, alongside top teams from FEU Alabang and Cebu Institute of Technology.

QS Reimagine Education Awards 2025 | *Top 20% Globally***October 2025**

- Submitted the Scaffl.ed intelligent tutoring system to the QS Reimagine Education Awards & Conference 2025 in London, UK, themed "Mind the Gap" and focused on higher education and AI integration.
- Shortlisted in the top 20% for the Best Use of Generative AI Award out of 1,600+ global submissions evaluated by 800+ international judges across 18 award categories.

Byte Forward Hackathon 2025 | *Top 10 Finalist***August 2025**

- Competed in a nationwide 24-hour hackathon organized by Converge ICT, PCCI, and Rev21 Labs with support from DTI, DICT, and DOST, focused on building digital solutions for local MSMEs.
- Built and pitched the Pasada prototype (voice-assisted jeepney driver optimization PWA) at the NCR regional leg in Quezon City, advancing to the Top 10 Finalists.

Create & Conquer 2025 Hackathon | *ZenithPads Award***July 2025**

- Advanced from 32 teams to the Top 9 Finalists at Create & Conquer 2025, an inaugural hackathon hosted by CpEO at FEU Tech.

- Won the ZenithPads Award for the Vyve project (AI-powered freelancer productivity overlay) and pitched to a panel from DOST-NCR, PwC Philippines, iCorePay, and DICT.

SAS Curiosity Cup 2025 | *Global Rank 22 / 112*

May 2025

- Analyzed the correlation between global food prices and life expectancy using WFP historical data with Python (NumPy, Pandas) and SAS in a global data analytics competition organized by SAS Institute.
- Scored 61/80 points, ranking 22nd globally out of 112 teams from 29 countries, placing in the top 20% internationally.

AppCon 2024 | *Top 90 / 178 Nationwide*

May 2025

- Submitted the Scaff.ed intelligent tutoring system to AppCon 2024, a nationwide application development competition organized by OTis Philippines Inc. with the theme "AI and IoT Solutions for Social Issues in the Philippines."
- Advanced as 1 of 90 qualifiers from 178 project submissions across 79 colleges and universities nationwide; one of only two UE CCSS teams to qualify.

2025 National Innovation Day | *Grand Prize (Public Administration category)*

April 2025

- Built and pitched the GabAI project (SMS-based AI chatbot for rural barangay governance) at the HABI: Filipinnovation Workshop organized by the National Innovation Council, NEDA, DOST, DTI, and CHED.
- Won the Grand Prize in the Public Administration category, competing against 55 student innovation teams from 62 Higher Education Institutions nationwide at Novotel Manila Araneta City.

CodeChum National Programming Challenge 2024 | *4th Place*

December 2024

- Won 1st Place in Group Stage and 1st Place in Semi-Finals before placing 3rd in the Finals and 4th in the Grand Finals as one of the top competitive programming teams in the country.
- Competed against 20+ partner institutions nationwide in timed algorithmic challenges, alongside top finishers Cebu Institute of Technology, Philippine Science High School, and Mapua University.

13th UMak IT Skills Olympics | *1st Runner-Up*

November 2024

- Awarded 1st Runner-Up in the IT Quiz Bee category at the 13th UMak IT Skills Olympics, competing against 60+ colleges and universities nationwide.
- Tested on comprehensive computing fundamentals spanning OOP, algorithms, networking, database systems, OS principles, and cybersecurity.

Tagisan ng Talino 2024 | *4th Place*

October 2024

- Represented the College of Computer Studies and Systems (CCSS) in UE Manila's inter-college academic competition, competing in the Quiz Bee covering Mathematics, Science & Technology, and General Information.
- Placed 4th in the Quiz Bee category, contributing to CCSS finishing 3rd overall in the combined inter-college standings.

StartUpQC | *Participant*

October 2024

- Submitted the Studdy project to the Quezon City Government's collegiate startup competition focused on high-impact technology solutions.
- Competed among 30+ teams from 14+ schools in the NCR across EdTech, AgriTech, GreenTech, and GovTech categories.

Philippine Startup Challenge 9 | *2nd Runner-Up, Best Startup Logo, Best Pitch Presentation*

October 2024

- Advanced from 50+ entries to the Top 15 Finalists at PSC9, DICT's flagship national startup program, and pitched the Studdy project at the NCR Regional Competition at QCX Business Center.
- Won 2nd Runner-Up, Best Startup Logo, and Best Pitch Presentation at the NCR Regional Pitching Competition.

12th UMak IT Skills Olympics | *1st Runner-Up*

November 2023

- Competed in the Java Programming category at the 12th UMak IT Skills Olympics, solving timed programming challenges covering sorting, ciphering, and data structures as a team of two.
- Awarded 1st Runner-Up among 50+ schools nationwide in the Java Programming category.

Philippine Startup Challenge 8 | *Top 27 NCR Finalists*

October 2023

- Developed and pitched a tech-driven solution at PSC8, DICT's flagship national startup program, as a first university-level startup competition experience.
- Shortlisted as one of the Top 27 NCR Regional Finalists from all NCR submissions.

RESEARCH & PUBLICATIONS

Scaff.ed: A Multimodal, LLM-Driven Intelligent Tutoring System for Java Debugging Using Gaze and Mouse Interactions

March 2026

- Presented at the 2026 International Conference on Advances in Artificial Intelligence and Machine Learning (AAIML 2026) in Tokyo, Japan. ([DOI: 10.1109/AAIML67890.2026.11498122](https://doi.org/10.1109/AAIML67890.2026.11498122)).

CERTIFICATIONS

Information Technology Specialist (Python) | *Certiport*

October 2024